

Live sizing-gate lockout, order-fill gap, and telemetry divergence

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Scope: System-2 execution engine (VM) · System-3 Layer-5 sizing/guardian · `gs://scalable-brain-artifacts/telemetry/`

Evidence window: 2026-07-14 → 2026-07-22 11:14 UTC · **Account:** OANDA practice `101-002-38449021-001` (CAD)

Status: Proposed — observational, no code changed

The trading loop is running and producing signals, but has not opened a position in over 24 hours and has not booked a profitable trade ever. Two independent gates in the Layer-5 sizing stage are now in a state that is *arithmetically unsatisfiable*, and the seven orders the system did approve today produced zero fills. Meanwhile the telemetry object most likely to be read by a dashboard has been frozen since 2026-07-18 and reports the system as dead.

1. Executive summary

Five findings, ordered by severity. Findings 1–3 are live and blocking; finding 4 is a historical defect corrected by a restart this morning; finding 5 is an observability failure that hid all of the above.

- 1. The sizing gate can never open again.** `min_live_trades = 20` is evaluated against a live-stats window capped at `168` hours. The system realizes roughly one trade per day, so at most ~7 trades can ever exist inside a 7-day window. The counter has been stuck at 9 since 07-20 and, as of 10:00 UTC today, has begun failing the *staleness* check as well. This is divergent: the longer it waits, the further it gets from the threshold.
- 2. Seven approved orders today produced zero trades.** System-3 reports `orders_published_total: 7`, yet the day ledger shows `trades: []` and OANDA reports `openTradeCount: 0`. Orders are leaving the system and not becoming positions.
- 3. System-2 and System-3 disagree about open positions.** S3 state says `open_positions: 1`; S2, and the broker itself, say zero. The `reconciliation_divergence` counter stands at 6. S3's drawdown and exposure maths are running on a position that does not exist.
- 4. Sizing ran in the wrong account currency for at least three days.** Every decision from 07-20 through 07-22 02:00 UTC computed against `account_ccy: "USD"` on an account denominated in **CAD**. A restart at ~02:25 UTC today corrected it. This is the live recurrence of the class of defect already filed as FIX-S3-004.
- 5. The published telemetry aggregate is four days stale and reports null health.** `telemetry/latest.json` froze at 2026-07-18T22:50Z with every S2/S3 payload `null`, while `telemetry/latest-vm.json` updates every few minutes with complete data. Any consumer reading the conventional filename sees a dead system.

FINANCIAL CONTEXT

10 realized trades, **all losers**: profit factor `0.0`, expectancy **-367.37 CAD/trade**, 30-day Sharpe `-11.49`, cumulative **-3,673.68 CAD** over the window. Lifetime account P/L is **-15,934.81 CAD**. Guardian state is **CAUTION** at 4.83% drawdown from peak.

The gates described below are currently the *only* thing preventing further loss. Fixing them without first addressing why every trade loses would convert a stalled system into a reliably losing one. **Finding 1 should not be "fixed" by lowering the threshold.**

2. Evidence

2.1 Daily ledger

From `telemetry/history/index.json` and the per-day decision files:

Date	Decisions	Approved	Trades	P/L (CAD)	Note
2026-07-14	0	—	—	—	day file empty; absent from index
2026-07-15	121	13	6	-2,807.75	rejects mostly <code>max_concurrent</code>
2026-07-16	152	0	0	0.00	
2026-07-17	125	0	0	0.00	
2026-07-18	0	—	—	—	outage; <code>latest.json</code> freezes 22:50Z
2026-07-19	17	4	3	-664.20	partial day
2026-07-20	192	0	0	0.00	168 × <code>insufficient_live_stats</code>
2026-07-21	193	1	1	-201.73	168 × <code>insufficient_live_stats</code>
2026-07-22	96	7	0	0.00	partial; 7 approved, 0 filled
Total	896	25	10	-3,673.68	-367.37/trade

2.2 Rejection reasons, 2026-07-22 (96 decisions)

Outcome	Reason	Count	Account ccy
REJECTED	<code>insufficient_live_stats</code>	63	CAD
REJECTED	<code>insufficient_live_stats</code>	21	USD
REDUCED	—	7	CAD
REJECTED	<code>no_conversion_rate</code>	3	USD
REJECTED	<code>stale_live_stats</code>	2	CAD

All 96 `signal_id` values are distinct — there is one evaluator, not two. Every rejection is at `layer: "sizing"`.

3. Finding 1 — the sizing gate is arithmetically unsatisfiable

P0 · LIVE · BLOCKING

To size a position the gate requires `live_trades ≥ min_live_trades (20)`, drawn from a live-stats window whose maximum age is `168` hours. Realized throughput is ~ 1 trade/day. A 168-hour window therefore cannot hold more than ~ 7 trades — the threshold of 20 is **unreachable by construction**.

Observed rejection payload (07-22 00:00:25Z, EUR_USD):

```
{"outcome": "REJECTED", "layer": "sizing", "reason": "insufficient_live_stats",  
"sizing": {"live_trades": 9, "min_live_trades": 20, "pair": "EUR_USD", "account_ccy": "USD"}}
```

The `live_trades` counter has been pinned at **9** since 2026-07-20 (it read 7 on 07-19, 0 on 07-15). It has not moved in three days because the only mechanism that increments it — closing a trade — is itself gated behind the counter. That is a closed loop.

As of **10:00 UTC today** a second, independent gate began firing on the same population:

```
{"outcome": "REJECTED", "reason": "stale_live_stats",  
"sizing": {"stats_age_hours": 168.28, "max": 168.0, "pair": "USD_JPY"}}
```

The stats backing those 9 trades have now aged past the 168-hour cap. This is the important part: the two gates are not merely both failing, they are **moving in opposite directions**. Trades age out of the window faster than new ones enter it, so `live_trades` will decline from 9 toward 0 while the threshold stays at 20. Left alone, the system will never trade again.

Recommended direction

1. Treat this as a *bootstrapping* problem, not a threshold problem. A live-stats requirement that a cold system cannot satisfy needs an explicit warm-up path — e.g. seed the window from backtested or paper statistics with a provenance flag, and require the flag to clear before real capital.
2. Make the two parameters mutually consistent by construction. Assert at startup that `min_live_trades` is achievable within `max_stats_age_hours` at the observed approval rate, and fail loudly if not. A gate that cannot open is a configuration error, not a risk control.
3. Emit a distinct telemetry signal for "gate is structurally closed" so this surfaces in minutes rather than days.

DO NOT SIMPLY LOWER THE THRESHOLD

Given profit factor `0.0` across 10 trades, opening this gate re-enables a strategy with no demonstrated live edge. The gate is doing accidental good. Sequence the strategy question first.

4. Finding 2 — approved orders are not becoming trades

P0 · LIVE · SILENT

System-3 counters: `decisions_approved_total: 7`, `orders_published_total: 7`. The 07-22 day file:
`"trades": []`. OANDA account summary: `openTradeCount: 0`, `openPositionCount: 0`,
`pendingOrderCount: 0`.

Seven orders were published today and none of them exists at the broker in any state — not open, not pending, not recorded as a closed trade. Either publication is not reaching the broker adapter, or fills are returning and being discarded. This is the same failure surface as the already-filed **FIX-S2-002** (live OANDA fills recorded as FAILED under a broker return-contract mismatch); this note is evidence that it is still live, or has recurred.

Note the compounding effect with Finding 1: orders that never become trades can never increment `live_trades`, so Finding 2 is one of the reasons Finding 1 cannot self-resolve.

5. Finding 3 — position-state divergence between S2 and S3

P1 · LIVE

```
s3state.open_positions: 1 versus s2status.open_positions: [], positions.count: 0, and
broker openTradeCount: 0. The S3 counter reconciliation_divergence reads 6 against
snapshots_reconciled: 529.
```

The Guardian is computing drawdown, exposure and state transitions against a phantom position. Its current verdict — **CAUTION** at 4.83% drawdown — is therefore not trustworthy, and neither are any correlation or exposure gates that read position state (cf. FIX-S3-001, which found those gates blind to open positions).

Note also that `peak_equity` is reported as `88,294.53`, but the 07-2026 equity series only begins at 2026-07-19T18:05Z with a maximum of `84,898.99`. The peak is carried from outside the retained series, so drawdown percentage cannot be reproduced from published telemetry alone.

6. Finding 4 — sizing ran in USD against a CAD account

P1 · HISTORICAL · CORRECTED BY RESTART

Hour-by-hour `account_ccy` in the sizing payload: **USD** for all 192 decisions on 07-20, all 193 on 07-21, and 07-22 00:00–02:00; **CAD** from 07-22 03:00 onward. S2 uptime places the process start at **~02:25 UTC on 07-22**.

The account is denominated in CAD (`account_summary.currency: "CAD"`). For three days the risk cap, Kelly fraction and unit conversion were computed against the wrong denomination. The visible symptom was `no_conversion_rate:`

```
"conversion_error": "no conversion from JPY to USD: need 'JPY_USD' or 'USD_JPY' in snapshot
prices"
```

— which is doubly odd, since `USD_JPY` is one of the eight pairs the signal producer watches. Both symptoms disappeared at the 03:00 restart boundary. This is not currently misbehaving, but it went unnoticed for three days and there is no assertion tying sizing currency to the broker's reported account currency. It is the live instance of **FIX-S3-004**.

Recommended direction

- Read account currency from the broker summary at startup and on every reconnect; never default it.

- Refuse to size — loudly — when the sizing currency does not match the broker's reported currency, rather than proceeding and failing later at conversion.
- Add the observed currency to a startup banner and to `s2status`, so a mismatch is visible without reading individual decision payloads.

7. Finding 5 — the published telemetry aggregate is stale and reports null

P1 • OBSERVABILITY

Two objects, two publishers, one convention-shaped filename:

Object	Published	Payloads
<code>telemetry/latest-vm.json</code>	2026-07-22T11:14Z (live)	all 8 populated
<code>telemetry/latest.json</code>	2026-07-18T22:50Z (4d stale)	<code>s2status</code> , <code>s2health</code> , <code>s2regime</code> , <code>s2signal</code> , <code>s3health</code> , <code>s3state</code> all <code>null</code> ; only <code>strategy</code> present

Anything consuming `latest.json` — the obvious default — has shown a system with no health, no regime, no signals and no state since 07-18. That is precisely the window in which findings 1 through 4 developed. The freeze timestamp also coincides with the 07-18 outage day, suggesting the aggregating publisher died then and was never restarted, while the VM-side publisher recovered.

Two smaller observability gaps in the same area:

- **Counter disagreement.** `s2signal.signals_produced_today`: 72 with `last_signal_at` 11:00Z, but `s2status.queue.messages_seen`: 7 with `last_message_at` 09:00:58Z and `staleness_sec`: 8005 (2.2 h). The producer's own ledger shows 07-20 as 192 evaluations / 192 approved / **16 published**. Signal delivery appears lossy, and the day file's `signals.aggregates` (96/96/96) agrees with none of them.
- **No Cloud Monitoring.** The project `scalable-brain` has zero Monitoring dashboards and zero alert coverage; all telemetry is hand-rolled JSON in a bucket. Nothing would have paged on a four-day publish freeze.

8. Cross-cutting observation: the running code is not in any checkout

Searching every local repository — `scalable-brain/`, `OtherSystems/system-2-execution-engine/`, `system-3-account-management/`, `deployment-guide/` — for the identifiers that appear in live telemetry returns **no matches**:

```
grep -rn "insufficient_live_stats|min_live_trades|latest-vm" → 0 hits
```

The Layer-5 sizing gate and the VM telemetry publisher exist only on the VM. There is no version-controlled source for the code currently making live sizing decisions, which means none of the fixes above can be reviewed, tested, or rolled back through the normal path. **This should be resolved before any of the fixes are attempted**, or the fix itself becomes another unversioned VM edit.

9. Suggested sequencing

1. **Get the code under version control.** Nothing else is safely actionable until the VM's sizing and telemetry code is in a repository (§8).
2. **Restore observability.** Repoint consumers at `latest-vm.json` or restart the aggregating publisher, then reconcile the three disagreeing signal counters (§7). Without this, no fix can be confirmed to have worked.
3. **Fix the order → fill gap** (§4). This is the highest-value repair: it is silent, it loses real orders, and it starves the very counter that Finding 1 is blocked on.
4. **Reconcile position state** (§5), so the Guardian's drawdown and exposure numbers mean something.
5. **Add the currency assertion** (§6). Cheap, and prevents a three-day silent recurrence.
6. **Only then** address the sizing gate (§3) — and address it as a warm-up/bootstrapping design question, alongside the open strategy question of why all 10 realized trades lost.

10. Reproduction

All evidence is readable with the `system1-rw` service account, which has access to the artifacts bucket only (Monitoring, Logging, Cloud Run and bucket listing are all denied):

```
gsutil cat gs://scalable-brain-artifacts/telemetry/latest-vm.json
gsutil cat gs://scalable-brain-artifacts/telemetry/latest.json
gsutil cat gs://scalable-brain-artifacts/telemetry/history/index.json
gsutil cp gs://scalable-brain-artifacts/telemetry/history/day/2026-07-2*.json .
```

Prepared 2026-07-22 from live telemetry at `gs://scalable-brain-artifacts/telemetry/`, snapshot 11:14:24Z. All figures are as-published; no code was modified and no live state was touched. Related register entries: FIX-S2-002, FIX-S3-001, FIX-S3-003, FIX-S3-004.